

```
In [1]: import pandas as pd
```

```
In [2]: pd.__version__
```

```
Out[2]: '2.3.3'
```

```
In [6]: import pandas as pd

df = pd.read_excel(
    r'C:\Users\polin\Desktop\GenAI_AgenticAI_Prac\Python Prac\data.xlsx'
)
```

```
In [12]: df
```

```
Out[12]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [13]: df.shape # rows & column from the dataframe
```

```
Out[13]: (195, 5)
```

```
In [14]: len(df)
```

```
Out[14]: 195
```

```
In [15]: df.columns
```

```
Out[15]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
              'IncomeGroup'],  
              dtype='object')
```

```
In [16]: len(df.columns)
```

```
Out[16]: 5
```

```
In [17]: df
```

```
Out[17]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [18]: df.isnull()
```

Out[18]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [19]: `df.isnull().sum()`

Out[19]: CountryName 0
CountryCode 0
BirthRate 0
InternetUsers 0
IncomeGroup 0
dtype: int64

In [20]: `df.isna().sum()`

Out[20]: CountryName 0
CountryCode 0
BirthRate 0
InternetUsers 0
IncomeGroup 0
dtype: int64

In [21]: `type(df)`

Out[21]: `pandas.core.frame.DataFrame`

In [22]: `df.info()` *#information of the dataframe*

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB

```

In [23]: `df.head()`

Out[23]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [24]: `df.head(10)`

Out[24]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [25]: `df.tail()`

```
Out[25]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

```
In [26]: df.tail(2)
```

```
Out[26]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

```
In [27]: df[:] # slicing in pandas
```

```
Out[27]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [28]: df[10:20]
```

Out[28]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Azerbaijan	AZE	18.300	58.70000	Upper middle income
11	Burundi	BDI	44.151	1.30000	Low income
12	Belgium	BEL	11.200	82.17020	High income
13	Benin	BEN	36.440	4.90000	Low income
14	Burkina Faso	BFA	40.551	9.10000	Low income
15	Bangladesh	BGD	20.142	6.63000	Lower middle income
16	Bulgaria	BGR	9.200	53.06150	Upper middle income
17	Bahrain	BHR	15.040	90.00004	High income
18	Bahamas, The	BHS	15.339	72.00000	High income
19	Bosnia and Herzegovina	BIH	9.062	57.79000	Upper middle income

In [29]: `df[:: -1]`

Out[29]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [30]: `df[0:200:15]`

Out[30]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
15	Bangladesh	BGD	20.142	6.6300	Lower middle income
30	Canada	CAN	10.900	85.8000	High income
45	Germany	DEU	8.500	84.1700	High income
60	Gabon	GAB	30.555	9.2000	Upper middle income
75	Honduras	HND	21.593	17.8000	Lower middle income
90	Kazakhstan	KAZ	22.730	54.0000	Upper middle income
105	Lithuania	LTU	10.100	68.4529	High income
120	Mozambique	MOZ	39.705	5.4000	Low income
135	Pakistan	PAK	29.582	10.9000	Lower middle income
150	Sudan	SDN	33.477	22.7000	Lower middle income
165	Seychelles	SYC	18.600	50.4000	High income
180	Uruguay	URY	14.374	57.6900	High income

In [31]: `df.columns`

Out[31]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup'], dtype='object')

In [32]: `df['CountryName']`

Out[32]:

```

0           Aruba
1    Afghanistan
2           Angola
3          Albania
4    United Arab Emirates
...
190        Yemen, Rep.
191        South Africa
192    Congo, Dem. Rep.
193           Zambia
194          Zimbabwe
Name: CountryName, Length: 195, dtype: object

```

In [33]: `df['CountryCode']`

```
Out[33]: 0      ABW
1      AFG
2      AGO
3      ALB
4      ARE
...
190    YEM
191    ZAF
192    COD
193    ZMB
194    ZWE
Name: CountryCode, Length: 195, dtype: object
```

```
In [34]: df[['CountryName', 'CountryCode']]
```

```
Out[34]:
```

	CountryName	CountryCode
0	Aruba	ABW
1	Afghanistan	AFG
2	Angola	AGO
3	Albania	ALB
4	United Arab Emirates	ARE
...	...	...
190	Yemen, Rep.	YEM
191	South Africa	ZAF
192	Congo, Dem. Rep.	COD
193	Zambia	ZMB
194	Zimbabwe	ZWE

195 rows × 2 columns

```
In [35]: # divide the dataset into categorical & numerical
```

```
In [36]: df.columns
```

```
Out[36]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
              'IncomeGroup'],
              dtype='object')
```

```
In [37]: df.head(1)
```

```
Out[37]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

```
In [38]: df_cat = df[['CountryName', 'CountryCode', 'IncomeGroup']]
df_cat
```

Out[38]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

```
In [39]: df_num = df[['BirthRate', 'InternetUsers']]
df_num
```

Out[39]:

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
...	...	...
190	32.947	20.0
191	20.850	46.5
192	42.394	2.2
193	40.471	15.4
194	35.715	18.5

195 rows × 2 columns

```
In [40]: print(len(df.columns))
print(len(df_cat.columns))
print(len(df_num.columns))
```

5
3
2

```
In [41]: print(df.shape)
print(df_cat.shape)
print(df_num.shape)
```

```
(195, 5)
(195, 3)
(195, 2)
```

```
In [42]: df.describe()
```

```
Out[42]:
```

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

```
In [43]: df_cat.head()
```

```
Out[43]:
```

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income

```
In [44]: df_cat.describe()
```

```
Out[44]:
```

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

```
In [45]: df_num.describe()
```

Out[45]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [46]: `df.columns`

Out[46]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup'], dtype='object')

In [47]: `df.columns = ['a', 'b', 'c', 'd', 'e']`

In [48]: `df.columns`

Out[48]: Index(['a', 'b', 'c', 'd', 'e'], dtype='object')

In [49]: `df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup']`

In [50]: `df.head(1)`

Out[50]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

In [51]: `df.BirthRate * df.InternetUsers`

Out[51]:

0	808.2516
1	207.9927
2	878.3135
3	736.5644
4	971.8720
...	
190	658.9400
191	969.5250
192	93.2668
193	623.2534
194	660.7275

Length: 195, dtype: float64

In [52]: `df['myCalc'] = df.BirthRate * df.InternetUsers`

In [53]: `df.columns`

```
Out[53]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
              'IncomeGroup', 'myCalc'],  
              dtype='object')
```

```
In [54]: df.head()
```

```
Out[54]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	myCalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720

```
In [ ]: df = df.drop('myCalc',axis = 1)
```

```
In [ ]: df
```

```
In [ ]: df['IncomeGroup']
```

```
In [ ]: df[df.IncomeGroup == 'Low income']
```

```
In [ ]: df['IncomeGroup'].unique()
```

```
In [ ]: df['IncomeGroup'].nunique()
```